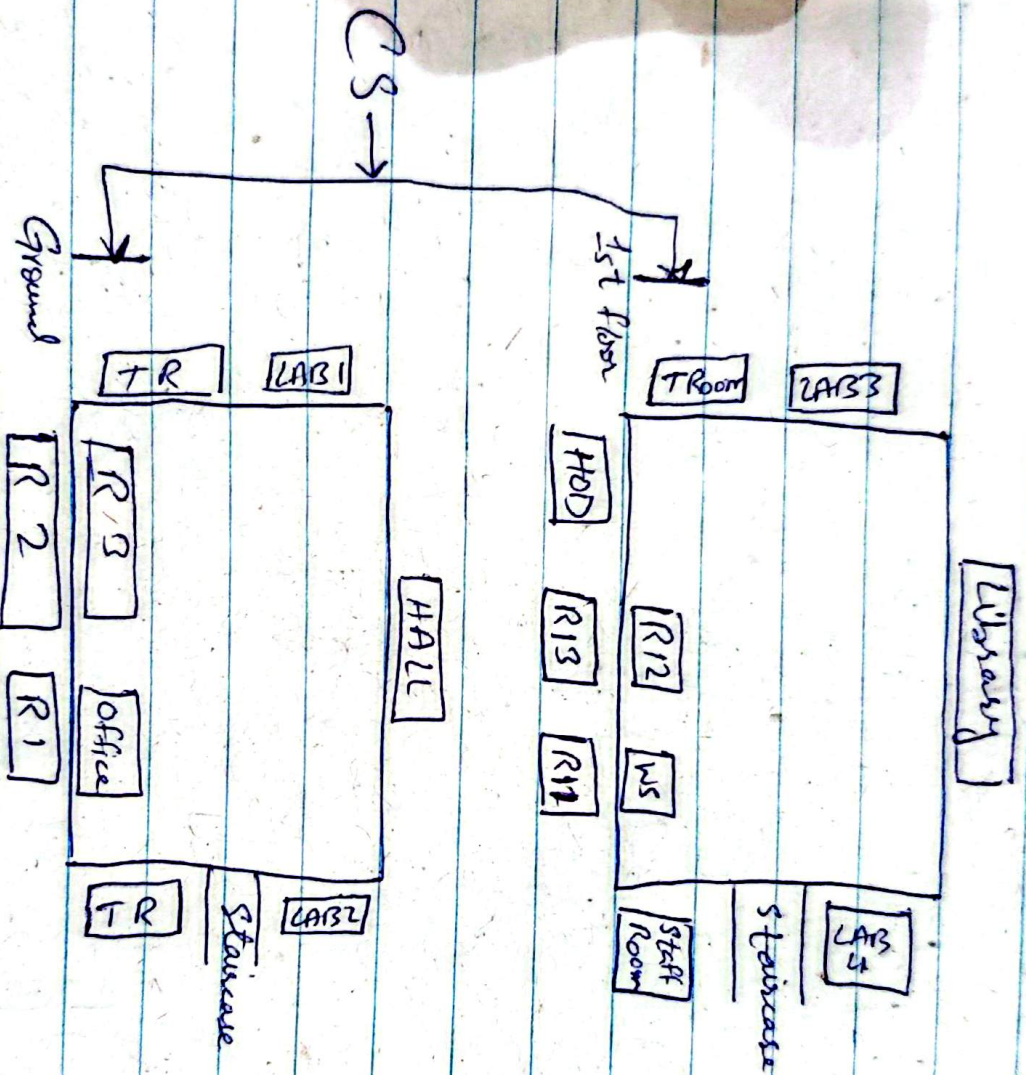


Map of CS Department



Formulation of Problem:

Initial state $\Rightarrow$ Library

Goal state $\Rightarrow$ Hall

State space $\Rightarrow$ All possible routes & location in CS.

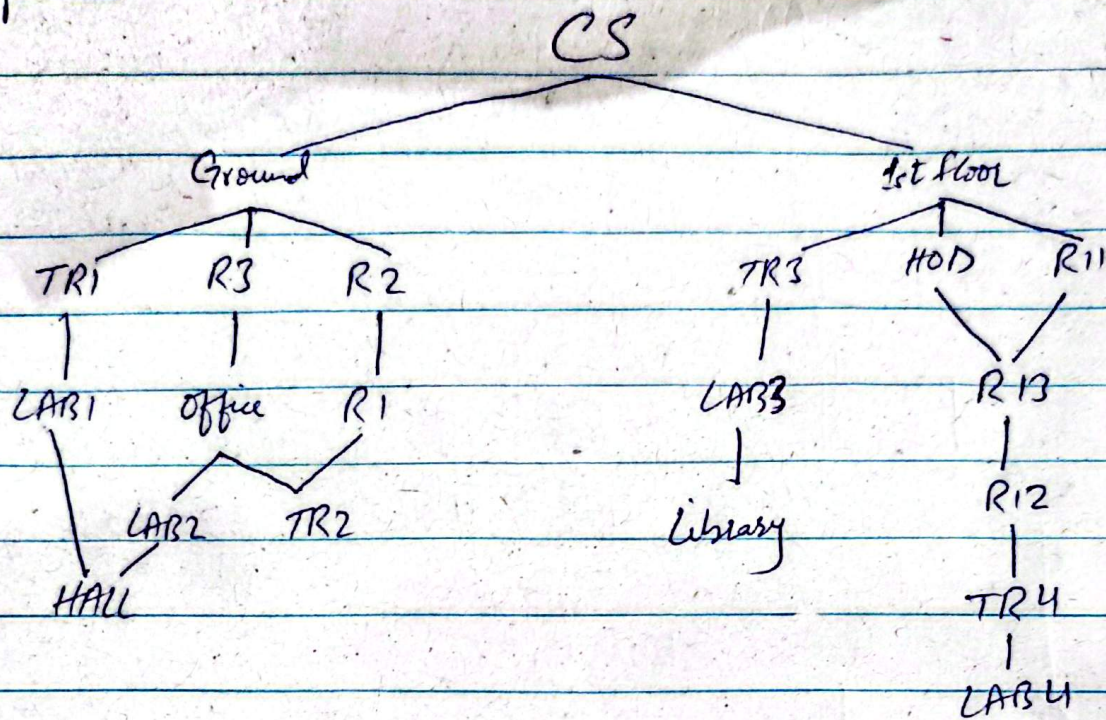
Action $\Rightarrow$ Move from one possible state to an

another state.

Transition Model $\Rightarrow$ Purpose to move state

Path Cost $\Rightarrow$ Distance.

Graph:-



BFS Table:-

Initialization:-

$Q = [\text{Library}]$

$V = [\text{Library}]$

deq library & expand

$Q = [\text{Lab3}]$

$V = [\text{Library}, \text{Lab3}]$

deq Lab3 & expand

$Q = [\text{TR3}]$

$V = [\text{Library}, \text{Lab3}, \text{TR3}]$

deq TR3 & expand

$Q = [\text{1st floor}]$

$V = [// //, \text{TR3}, \text{1st floor}]$

deq 1st floor & expand

$Q = \text{[CS, HOD, R11]}$

$V = \text{[// //, 1st floor, CS, HOD, R11]}$

deq CS & expand

$Q = \text{[HOD, R11, Ground]}$

$V = \text{[// //, R11, Ground]}$

deq HOD & expand

$Q = \text{[R11, Ground, R13]}$

$V = \text{[// //, Ground, R13]}$

deq R11 & expand

$Q = \text{[Ground, R13]}$

$V = \text{[// //, Ground, R13]}$

deq Ground & expand

$Q = \text{[R13, TRI, R3, R2]}$

$V = \text{[// //, R13, TRI, R3, R2]}$

deq R13 & expand

$Q = \text{[TRI, R3, R2, R12]}$

$V = \text{[// //, R12]}$

deq TRI & expand

$Q = \text{[R3, R2, ~~R12~~, R12, Lab1]}$

$V = \text{[// //, Lab1]}$

deq R3, R2, R1, Lab1 & expand

$Q = \text{[Hall]}$

$V = \text{[// //, R3, R2, R12, LAB1, Hall]}$

Shortest Path Library To Hall

Library



LAB3



TR3



1st floor



CS



Ground



TR1



LAB1



HALL